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unique stitching machine focus light, for which the company has filed a patent application.

Scope for customisation

LED lighting products have huge scope for customisation, starting from manufacturing and design of new casings, housings and enclosures to design of luminaires as per customer specifications. Even the fixtures and drivers are being customised.

"Manufacturers have been designing and making products that suit their customers well, by not just optimising lighting levels that can be achieved through customisation of colour temperatures, CRI, lux levels and the axis but also providing optimal light distribution by using specialised optical lenses," says Sinha and Tyagi.

"But while customising products, the manufacturer should always keep quality as the priority, as in this competitive market, manufacturers end up using inferior-quality components that not only hamper the efficient operation of the product but also adversely impact the expected life of LEDs," adds Jindal.

According to Gurinder Singh Lali, founder, Nano Power, "Electronic ballasts are also responsible for better lamp performance, extending life and improving colour characteristics. Luminaires are selected for their lighting effectiveness, and these products are manufactured and customised as per specifications provided by the buyers, which include many factors like input power supply (AC/DC), lumens output (high/low), total wattage, angles of light output, and specifications of size and shape."

Technological changes and innovations

An LED lighting product is a combination of electronic design (LEDs), electrical design (LED drivers), mechanical design (cooling devices) and optical design (lamp holders), and innovations have been taking place in all these components. In the last two years, the performance of high-power LEDs has improved drastically, with greatly improved light output (flux), efficacy (lumens per watt), colour qual-



Avni Energy's downlighter

ity and stability.

"Today, commercial LEDs emit 110-120 lumens/watt and, recently, the Bilkent University in Turkey has claimed to have developed an LED with an output of 300 lumens/watts," says Joneja.

Some of the improvements in the last few years have given LED technology a real boost—ceramic packages for better thermal management, increase in efficiency within a short time frame backed by chip technologies, long-life, high-brightness white LEDs with longer maintenance intervals and better total cost of ownership. "These improvements support fast and extensive adoption of LEDs for different applications," opines Sood.

As the lumens/watt increases, the number of LEDs to be used to replace conventional lighting is coming down, thereby reducing the cost and wattage of replacing conventional lights.

Jindal shares, "Innovations in commercial lighting are more focused on designing the components for maximum light with minimal wastage. As a result, the angles of light dispersion are important



Avni's high bay light